RIG: MLLMS-GUIDED LAYOUT PLANNING FOR ZERO-SHOT MULTI-REFERENCE PERSONALIZATION

Junhao Feng *Zihao Zhang* *Xinghang Xu* *Zhonghua Wan*†

School of Mechanical Engineering, North University of China, Taiyuan, China

ABSTRACT

Though zero-shot adapters excel in image personalization, they often encounter significant challenges in multi-reference personalized generation, specifically failing to precisely adhere to the spatial layouts described in text prompts and suffering from feature leakage between reference images. To address these two challenges, we propose RIG (Regional Image-prompt Generation), a novel training-free framework. For the first challenge, leveraging Multimodal Large Language Models (MLLMs), we introduce a layout planning binder. Leveraging Chain-of-Thought (CoT) reasoning, this module infers and generates precise global layouts from text prompts, while simultaneously binding reference images to their corresponding regions. For the second, we introduce a satially decoupled diffusion mechanism that isolates feature streams during attention computation. By injecting reference features exclusively into designated regions, this mechanism effectively prevents feature interference between reference images. Extensive experiments demonstrate that RIG significantly outperforms state-of-the-art adapter methods in terms of both personalization fidelity and text-layout alignment.

Index Terms— Image Personalization, Diffusion Models, Multimodal LLMs, Zero-Shot Learning.

1. INTRODUCTION

Customizing or personalizing image generation with reference images is a significant and practical application of text-to-image diffusion models. In recent years, significant strides have been made in image personalization. Existing approaches generally fall into two paradigms: tuning-based or training-based and training-free methods. Although tuning-based methods (e.g., DreamBooth[1], Custom Diffusion[2]) or training-based methods (e.g., Mix-of-Show[3]) achieve high fidelity and strict prompt adherence via parameter fine-tuning or training, their high training costs and the necessity to manage task-specific parameters hinder rapid deployment in multi-task scenarios.

In contrast, training-free adapters (e.g., IP-Adapter[4], OminiControl[5]) achieve plug-and-play personalized generation by injecting reference features into frozen models, demonstrating exceptional performance in single-reference personalization tasks. However, existing models struggle to precisely reason about spatial relationships within text prompt, making it difficult for adapter-based methods to achieve personalized generation that aligns with the described spatial layouts(Fig. 1). The guidance from multiple reference images further overshadows the text prompts, hindering the model's ability to map specific reference images to their corresponding spatial locations as defined by the text.Furthermore, we observe that current adapters tend to broadcast extracted features across the entire generative latent space. Consequently, in multi-reference scenarios, features from a single image are erroneously applied as global guidance, failing to effectively disentangle the semantics of different reference images. This inability leads to conflicts between personalized attributes, manifesting as "feature leakage," where each generated subject exhibits visual characteristics blended from other reference images(Fig. 1).

Fig. 1. A comparison between RIG and previous adapter methods. The yellow and purple text are respectively the layout and background hints in the prompts.

To address the aforementioned two limitations, we present RIG(Regional Image-prompt Generation), a novel training-free framework for reference-guided image personalization. By translating the Multimodal Large Language Models(MLLMs)' region-level structured generation intents into the underlying control of the diffusion model, RIG explicitly defines the specific spatial region where each reference image should exert its influence. Within these confined regions, we employ spatially decoupled diffusion mechanism to achieve personalized generation.

The main contributions of this paper are as follows.

†Corresponding Author

• A novel training-free framework for personalized generation called RIG is presented. By seamlessly integrating the spatial reasoning capabilities of MLLMs with diffusion models, RIG achieves precise alignment between personalized content and textual spatial layout prompts without requiring training.

• A spatial decoupled diffusion mechanism is devised to enforce strict feature isolation during the attention computation phase, effectively addressing the persistent issues of feature leakage observed in adapter models for multi-reference personalized generation.

• Qualitative and quantitative experiments demonstrate the superiority of RIG. It significantly outperforms current existing advanced zero-shot adapters, particularly in terms of multi-reference and prompt-layout alignment.

2. RELATED WORKS

Training-free Personalized Image Generation Personalized generation aims to synthesize new images containing specific subjects or styles derived from a few reference images. To address efficiency bottlenecks, adapter-based methods like IP-Adapter have emerged. By employing a decoupled cross-attention mechanism, they inject reference features into frozen diffusion models. Although efficient, the original IP-Adapter relies on global feature injection, which often leads to attribute mixing and spatial layout error when extended to multi-reference scenarios. Although recent work like Conceptrol[6] attempts to guide attention using text-derived masks, it still fundamentally relies on the base model's weak spatial grounding and fails to guarantee strict feature isolation. In contrast, RIG eliminates global interference by enforcing explicit region-aware feature injection.

Controllable Image Generation To enhance structural control, methods such as ControlNet[7], T2I-Adapter[8], and GLIGEN[9] have been introduced. These allow users to guide generation using explicit spatial conditions like edge maps, segmentation masks, or bounding boxes. Although powerful, these approaches face two limitations in our context: (1) Dependency on explicit conditions: They require users to provide precise masks or coordinates, failing to support abstract natural language spatial instructions (e.g., "A is to the left of B"). (2) Lack of identity binding: They focus on generic object layout and do not address the challenge of binding specific reference identities to designated regions. RIG solves this by automating the semantic binding process directly from text.

MLLMs as Layout Planners Leveraging the reasoning capabilities of MLLMs to guide visual generation has become a burgeoning trend. Representative works like LMD[10] and RPG[11] utilize the in-context learning ability of LLMs(Large Language Models) to translate abstract prompts into concrete layout bounding boxes, guiding diffusion models to handle complex compositional scenarios. These methods demonstrate the potential of MLLMs in spatial planning. However, existing MLLM-based approaches are predominantly confined to generic text-to-image (T2I) generation tasks. They lack the mechanism to integrate specific reference image features into the planned regions.

Bridging these gaps, RIG uniquely integrates the "high-level semantic planning" of MLLMs with the "low-level feature injection" of adapters. Unlike previous works, we utilize the layout information parsed by the MLLM to directly modulate the cross-attention mechanism within the diffusion model. This enables simultaneous precise spatial control and high-fidelity multi-concept personalization without any fine-tuning.

3. METHOD

3.1. Framework Overview and Conception

This work aims to tackle the persistent issues faced by zero-shot adapters in multi-reference scenarios, specifically the inability to maintain precise spatial layout and prevent feature leakage. To this end, we propose the RIG framework, a novel zero-shot paradigm that eliminates the need for fine-tuning any pretrained models. RIG is composed of two core components: 1.Multimodal layout planning and binding(Fig. 2). 2.Spatially-decoupled diffusion(Fig. 3).

3.2. Multimodal Layout Planning and Binding

This module executes two primary tasks: Hierarchical layout decomposition and visual-semantic binding.

Given a global text prompt P containing multiple entities and their complex relationships, along with a set of reference images $IP = \{IP_1, \ldots, IP_j\}$. Traditional diffusion models struggle to process the intricate spatial relationships within P. We utilize an MLLM to deconstruct the image space $H * W$ at a semantic level. The planner infers and outputs a Generation Blueprint Φ formally defined as:

$$\Phi = (S, T, B) = \mathrm{MLLM}(P, IP) \tag{1}$$

This blueprint consists of three key components:

1.Spatial Partition Descriptor (S): To ensure the generated spatial layout adheres strictly to the textual description, we employ a custom template to enforce a Chain-of-Thought (CoT) reasoning process within the MLLM. The MLLM outputs a set of structured segmentation instructions (e.g., specifying the row and column proportions for each part), which we convert into a set of mutually exclusive binary regions: $M = \{M_0, \ldots, M_k\}$ where $M_k \in \{0, 1\}^{H \times W}$ defines the spatial extent of the k-th region.

2.Regional Text Prompts (T): User prompts are typically global, mixing attributes and relationships of various entities.

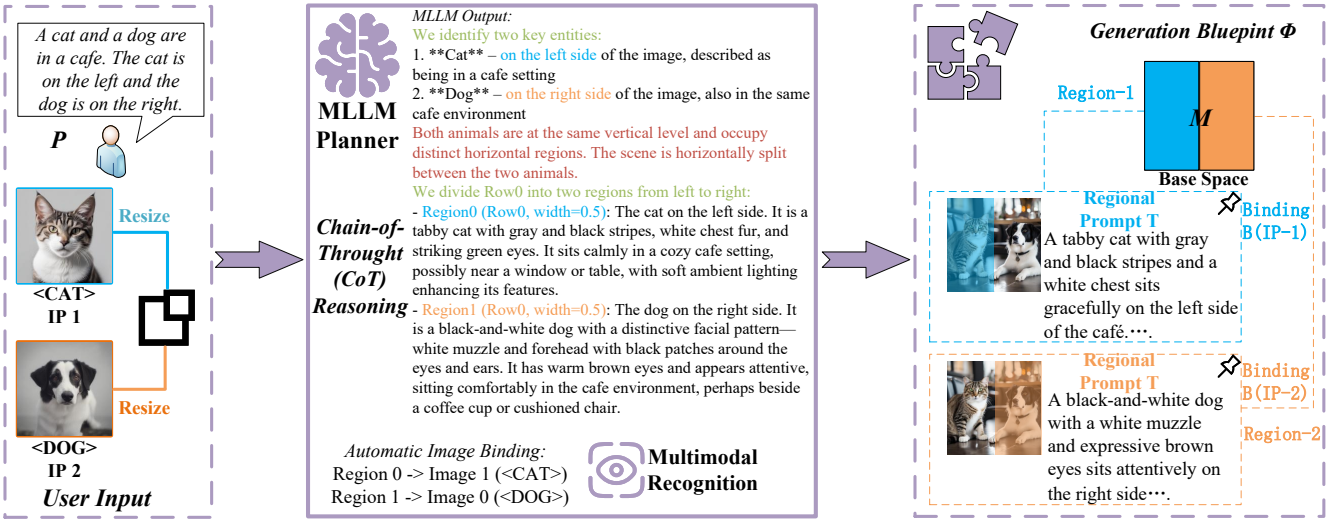

Fig. 2. Given a global prompt P and reference images IP_j, the MLLM performs chain-of-thought reasoning to decompose the scene. It outputs a generation blueprint Φ containing: (1) Spatial partition descriptor (S) represented by non-overlapping sub-regions M; (2) Regional text prompts (T) composed of fine-grained descriptions t_k; and (3) Visual-semantic bindings (B). The purple dashed lines illustrate the automated binding process, where specific regions are explicitly linked to reference images (e.g., the blue region binds to the CAT image IP_1).

For more effective regional guidance, the global prompt is rewritten into a set of fine-grained sub-prompts:

$$T = Redescribe\{t_0, \ldots, t_k\} \tag{2}$$

where $t_k$focuses on describing the local visual content of the k-th region. These descriptions are also derived by the MLLM based on our preset template.

3.Visual-Semantic Binding (B): Existing layout control methods (e.g., GLIGEN) often require users to manually specify image-location correspondences, limiting practicality. A core innovation of RIG is the introduction of Automated Visual-Reasoning Binding. The MLLM jointly analyzes the visual features of the reference images IP_j and the semantic content of the regional prompts t_k to construct a binding mapping function $B : \{0, \ldots, k\} \rightarrow \{-1, 1, \ldots, N\}$. If $(B(k) = j(j > 0))$, it indicates that the k-th region is explicitly bound to the reference image (e.g., binding the reference image "cat" to the left region). Additionally, we introduce a text-only control identifier (denoted as -1). If selected, this indicates that the region is controlled solely by text (e.g., background regions).

This mechanism (Fig. 2) ensures that the correspondence between reference images and fixed layout regions is logically locked prior to the commencement of the generation process.

3.3. Spatially Decoupled Diffusion

To precisely translate the generated blueprint into high-fidelity images, we propose the spatially decoupled diffusion mechanism built upon pretrained text-to-image models.

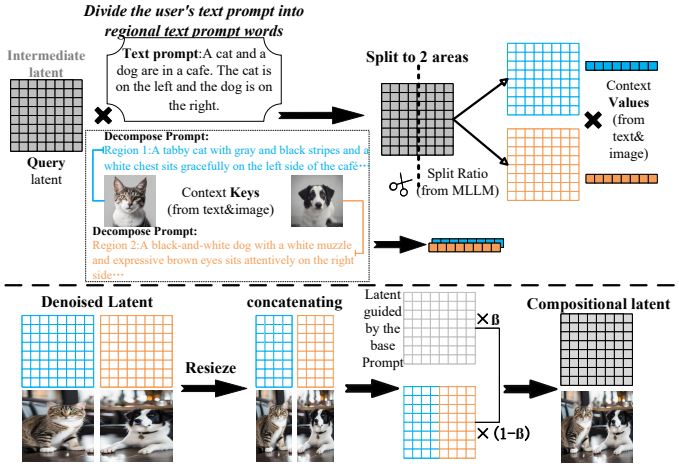

Fig. 3. Overview of the spatially decoupled diffusion architecture. These regional features are processed in parallel within isolated attention blocks to prevent feature leakage and are subsequently fused with the global base content to ensure overall coherence.

Leveraging the planned region M and binding mapping B, our method integrates into the Cross-Attention layers of the U-Net, employing a feature routing strategy to achieve disentangled control over feature generation(Fig. 3).

At the $(t-1)$-th denoising step of the diffusion process, let $z_{(t-1)} \in R^{h \times w \times c}$ denote the current latent feature map. To prevent feature interference between different subjects, we

first perform spatial decoupling on the Query vectors Q based on the downsampled regions M_k. Specifically, we compute the region-specific query $Q^{(k)}$ as:

$$Q^{(k)} = Q \odot M_k. \tag{3}$$

This operation spatially isolates the query features for the k-th region, ensuring they interact exclusively with the authorized conditional information, thereby fundamentally blocking feature confusion.

We did not consider each reference image as a global feature, we construct exclusive Key-Value pairs for each region k.The textual condition is derived from the local prompt t_k, while the visual condition comes from the bound reference image $IP_{B(k)}$. The attention output for the region is calculated as

$$\begin{aligned} O^{(k)} = & \underbrace{\mathrm{Attn}(Q^{(k)}, K_{txt}^{(k)}, V_{txt}^{(k)})}_{\text{Textual Control}} \\ & + \lambda \cdot IP_{[B(k)\neq -1]} \cdot \underbrace{\mathrm{Attn}(Q^{(k)}, K_{img}^{(k)}, V_{img}^{(k)})}_{\text{Visual Control}} \end{aligned} \tag{4}$$

where $K_{txt}^{(k)}$ and $V_{txt}^{(k)}$ are projected from the CLIP[12] embeddings of the region prompt t_k, and $K_{txt}^{(k)}$ and $V_{txt}^{(k)}$ are projected from the image encoder features of the reference image $IP_{B(k)}$. $I_{[.]}$ is an indicator function that activates the visual injection branch only when a valid reference image is bound to the region. λ serves as a scaling factor to modulate the influence weight of the reference image.

After computing the local attention outputs for all regions in parallel, we recompose them in the latent space. Since the regions are spatially mutually exclusive, the aggregated output O^{out} is obtained by summation:

$$O^{out} = \sum_{k=0}^{N} O^{(k)}. \tag{5}$$

However, direct concatenation of regional contents can lead to artificial segmentation boundaries or unnatural transitions. To mitigate this we fuse the aggregated output O^{out} with the global content O^{base}:

$$O_{final} = \beta \cdot O^{base} + (1-\beta) \cdot O^{out}. \tag{6}$$

This fusion mechanism, in conjunction with the original Self-Attention layers of the U-Net, ensuring consistency in background and style across different regions for seamless image generation.

4. EXPERIMENT

4.1. Experimental Details and Comparisons

We implement our scalable RIG framework using Qwen3-max[13] for spatial planning and Stable Diffusion XL[14] as the backbone (50 inference steps).

Table 1. Quantitative comparison on our collected RefBench. We evaluate 2D-Spatial, identity fidelity and text alignment. **Bold**: best results.(In the table, FT represents the method based on fine-tuning, and no-FT represents the method without fine-tuning)

Method	UnitDet ↑	CLIP-T ↑	CLIP-I ↑	Overall time (s) ↓
Textual Inversion (FT)[16]	0.157	0.693	0.673	>350
DreamBooth-LoRA (FT[1])	<u>0.164</u>	<u>0.718</u>	**0.692**	>350
ELITE (no-FT)[17]	0.128	0.657	0.633	<u>37</u>
IP-Adapter (Baseline/no-FT)[4]	0.134	0.645	0.629	**33**
Ours (Zero-shot)	**0.183**	**0.733**	<u>0.688</u>	39

In major personalized generation scenarios, we compare our framework against previous zero-shot adapters and several commercial closed-source methods[15]. The comparison primarily focuses on the following two aspects: 1.Personalized generation scenarios involving abstract spatial layouts and relational prompts. 2.Personalized generation scenarios with multi-reference image objects.

As shown in the figure(Fig. 4), the RIG framework significantly outperforms current zero-shot adapter methods on generation tasks with abstract spatial cues. In particular, it exhibits good visual and textual alignment in terms of the layout of generated content following textual representations, achieving similar performance to existing closed-source models that require finetuning and training. At the same time, in the scenario of multiple reference images generation, RIG well isolates the features between each reference image, and solves the challenge of reference image feature leakage in the scenario of multiple reference images personalized generation by the previous zero-shot adapter.

In Table 1.We re-implemented it according to the code and weights it released (the original evaluation datasets have not been made public). To assess the spatial-semantic alignment, we utilize the comprehensive T2I-CompBench++[18] benchmark, specifically focusing on 2D spatial relationships (UniDet). regarding personalized generation, we evaluate identity fidelity using CLIP-I[12] and text alignment using CLIP-T[12] (scaled by a factor of 2.5). Furthermore, we report the overall time (Generate the first image with RTX 4090D*1, 24G, Round off)to demonstrate the deployment efficiency of our framework.

4.2. Ablation Experiment

4.2.1. Ablation Test for Spatial Layout Planning

Spatial layout is essential for enforcing layout constraints. Its removal forces the model to rely exclusively on text prompts, often resulting in ambiguous compositions. By comparing our method with the base model[14] (Fig. 5), we demonstrate that this module provides the explicit guidance for the base model to achieve accurate and coherent image composition.

Fig. 4. Qualitatively comparisons of our method with the previous zero-shot adapter work and closed-source methods.

Fig. 5. A test of the impact of the MLLM planning on the composition of base model.

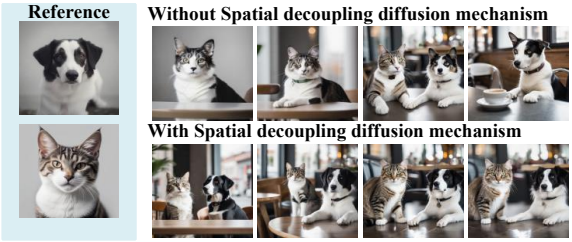

Fig. 6. The dissolution of the spatial decoupling diffusion mechanism.

4.2.2. Spatial Decoupling Diffusion Ablation Test

Conventional zero-shot adapters rely on global spatial decoupling, which inherently causes feature leakage in multi-subject scenarios due to the lack of instance-specific isolation. We conducted an ablation study using the cat-and-dog benchmark by retaining regional binding but removing our proposed spatial decoupling diffusion mechanism. As illustrated in Fig. 6, the absence of this mechanism resulted in severe global feature entanglement. This manifested as identity confusion or the blending of distinct subjects into a single hybrid entity, confirming the necessity of our approach for precise multi-subject generation.

5. CONCLUSION

In this paper, we introduce RIG - a framework that requires no training, which significantly enhances the zero-shot adapter method's compliance with spatial layout prompts in text and feature preservation between reference images in the task of image personalization generation. Our framework is based on observations of previous attention mechanisms and base pre-trained models: (1) The previous cross-attention mechanism decouples the global injection of image features. (2) The ambiguity of the native model for spatial layout prompts. These are the key factors that make it difficult to follow the spatial layout and feature leakage during the personalized generation process. Our RIG precisely starts from these perspectives, leveraging the powerful capabilities of multimodal large language models and our unique spatial decoupling diffusion mechanism to place reference images in the correct positions and isolate them for decoupling, achieving personalized tasks on the basis of aligning with text prompt words. This framework, which requires no training at all, offers a new paradigm for image personalization methods. Undoubtedly, it can be extended to more models and technologies, promoting the in-depth application of personalization generation tasks.

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